



HK IMO Training 2021-2022
Problem Set 22
For 19 March 2022



Problem 85. A positive integer is said to be *good* if it is 2 or it is of the form $3^m 5^n$ for some nonnegative integers m and n . Prove that every positive integer can be represented as the sum of different good integers.

Problem 86. Let $ABCD$ be a cyclic quadrilateral with circumcentre O . The diagonals AC and BD meet at E . Let O_1, O_2, O_3, O_4 be the circumcentres of $\triangle EAB, \triangle EBC, \triangle ECD, \triangle EDA$. Prove that O_1O_3 and O_2O_4 meet at the midpoint of OE .

Problem 87. X and Y play a game as follows. Initially, X picks a 3-digit number n from 000 to 999 (possibly starting with 0) secretly. In each round, Y chooses a 3-digit number m from 000 to 999. If m and n share at least one common digit in the same position, X will tell Y this is a *good* guess. Otherwise, X will tell Y this is a *bad* guess. For example, if X has chosen 012, then 012, 020 and 314 are good guesses, while 101, 123 and 456 are bad guesses. At least how many rounds are needed so that Y can guarantee to know the value of n ?

Problem 88. Let a_{ij} be the (i, j) th entry of an $n \times n$ matrix, where n is a positive integer. Suppose $a_{ij} > 0$ and $a_{ij}a_{ji} = 1$ for any i, j (possibly $i = j$). Find the maximum value of

$$\sum_{i=1}^n \frac{1}{\sum_{j=1}^n a_{ij}}.$$